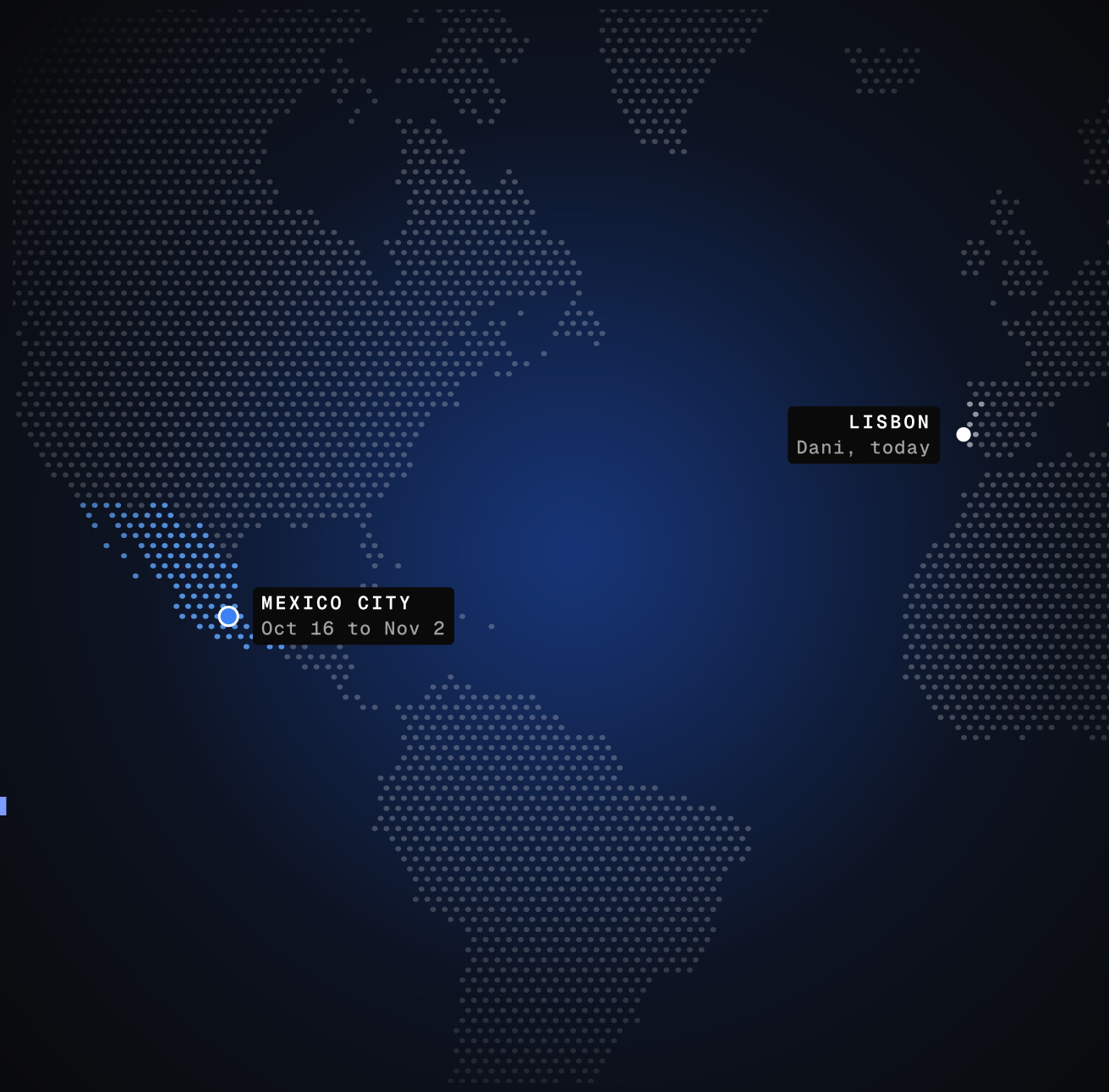
NOMAD.OS



ASSIGNMENT 5 · AI INTERACTION PROTOTYPE

Booked by AI. Approved by you.

A travel agent inside NOMAD.OS. It books trains, hotels, and dinner, and asks before it pays or shares anything.

IDSN 523 · Mitch Kostelecky and Ian Harris · Fall 2026

SPEAKER NOTES · 01 BOOKED BY AI. APPROVED BY YOU.

We’re presenting NOMAD.OS, a phone for people without a home base. For this assignment, we added one layer of AI: a travel agent inside the app. It books trains, hotels, and dinner from a chat, but it never pays or shares anything until you approve it. Our user, Dani, is in Lisbon and flies to Mexico City in three days.

EXISTING PRODUCT AND SELECTED EXPERIENCE

Airalo gets you online.
Then you're on your own.

30M+ travelers · **200+** destinations

SELECTED EXPERIENCE: THE THREE DAYS BEFORE A NEW COUNTRY

Airalo

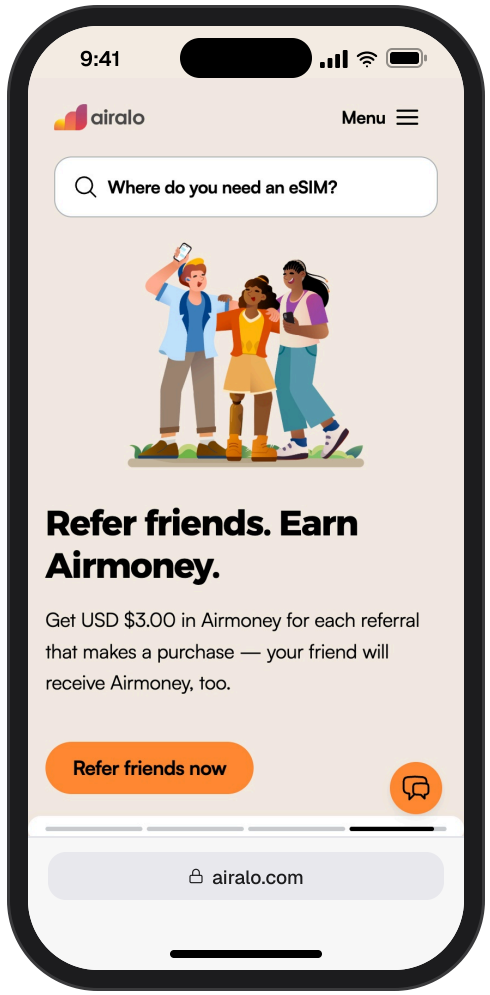
✓ Live in minutes ✓ No SIM swap ✓ Home line stays on

Pick a country → Buy data → Install → Connect → Top up

Everything else

× Done in other apps × Risks left to you

Stay · Train · Dinner · Codes and VPN



SPEAKER NOTES · 02 EXISTING PRODUCT: AIRALO

Our comparable product is Airalo, the largest eSIM marketplace, with over 30 million travelers. We focused on the three days before a new country. Airalo's five steps work well: it's live in minutes, and your home line stays on for bank codes. But it stops at data. The stay, the train, dinner, and bank and VPN decisions are left to other apps, and to you.

USER, GOAL, AND CURRENT FLOW

Dani lands in three days.
Every step is a different app.

Goal: land in Mexico City ready to work



Dani

Remote worker, moves every few weeks. A composite from our research.

Lisbon
Tue Oct 13

Buy an eSIM
Airalo
✓ Live in minutes

Pick a line for codes
Help articles
⚠ Advice conflicts
E14 E23

Book 17 nights
Booking site
⚠ Is the listing real?

Train, dinner, delivery
Three more apps
⚠ Card typed in again

Land and log in
Bank app
⚠ VPN freezes the bank
E13 E26

Mexico City
Fri Oct 16

SPEAKER NOTES · 03 USER, GOAL, AND CURRENT FLOW

Dani is a remote worker, a composite from our research, who wants to land in Mexico City ready to work. Buying the eSIM is easy. Every step after it is a different app with its own risk: conflicting advice about bank codes, a listing that might be fake, a card typed in again and again, and a VPN that can freeze the bank.

OPPORTUNITY STATEMENT

When **a remote worker** is trying to **land in a new country ready to work**, they experience **a pile of bookings and setup decisions spread across separate apps** because **each app sells one piece and leaves the risks in between to guesswork**. There is an opportunity to help them **book what they need and arrive set up, approving every payment and every detail shared**.

AI ISN'T AUTOMATICALLY THE ANSWER. WE COMPARED THREE OPTIONS.

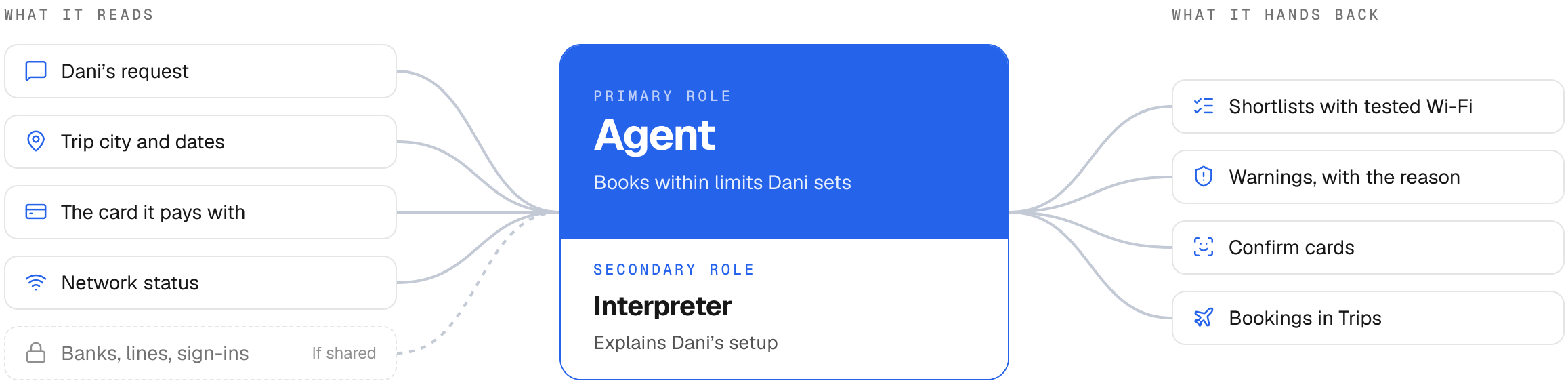
	A better checklist	A better booking site	An AI agent Our pick
Books it	✗	✓	✓
Fits Dani's card, network, and dates	✗	✗	✓
Flags risks between apps	✗	✗	✓
Explains each step	✓	✗	✓
	Kept as the fallback		Only with a confirm step and hard limits

SPEAKER NOTES · 04 OPPORTUNITY STATEMENT

Here's our opportunity. Remote workers juggle bookings and setup decisions across separate apps, and each app leaves the risks in between to guesswork. We want to help them book what they need and arrive set up, approving every payment and detail. AI isn't automatically the answer, so we compared three options. Only an agent books, fits your setup, spots risks, and explains each step, and only with a confirm step and hard limits.

AI ROLE AND RATIONALE

Primary role: agent.
Secondary role: interpreter.



Stays with Dani Choosing · Paying · What’s shared · Cancelling · Anything in a bank

SPEAKER NOTES · 05 AI ROLE AND RATIONALE

Our primary AI role is an agent. It takes actions for Dani, within limits Dani sets, because the safe choice depends on dates, card, and network all at once. Its secondary role is an interpreter that explains Dani’s setup. On the left is what it reads, and on the right is what it hands back. Choosing, paying, sharing, and cancelling always stay with Dani.

HUMAN AND AI RESPONSIBILITY MAP

The agent prepares.
Dani decides.

	Dani	Agent	Shared decision
Initiation	Asks, or taps a tile	Never starts on its own	Opens a suggested check
Input and interpretation	Says what's needed	Fills in city, dates, card	Missing details get asked
Generation or action	Compares the options	Shortlists and flags risks	Picking one
Decision or approval	**Confirms, changes, or cancels**	**Shows cost, card, what's shared**	**The confirm card**
Completion and review	Checks Trips, can cancel	Books and files it	Setup fixes: approve or skip

Asks first for money or personal details

Face ID over $100

Says why when it leaves something out

NOMAD.OS owns what the card showed

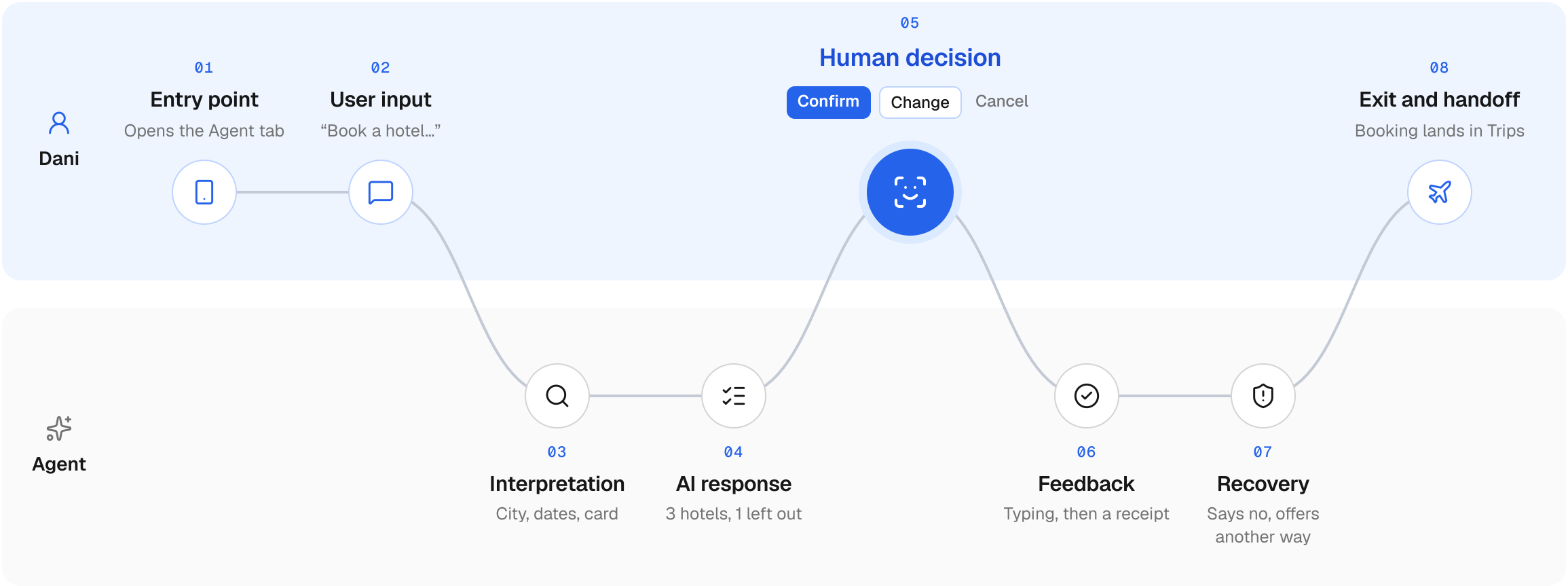
Never: pays off-site or by transfer, changes bank settings, hides location

SPEAKER NOTES · 06 RESPONSIBILITY MAP

In every row, the agent prepares and Dani decides. The agent never starts on its own. It fills in the details and shortlists the options. Dani picks one, and on the confirm card, Dani confirms, changes, or cancels. The agent always asks before spending money or sharing details, explains what it leaves out, and never touches bank settings. If it's wrong, we answer for what the card showed.

PROPOSED AI INTERACTION FLOW

Dani starts it and decides it.
The agent does the work, then hands it back.



SPEAKER NOTES · 07 INTERACTION FLOW

Here’s the flow over time, with Dani on top and the agent below. Dani starts it by opening the Agent tab and asking for a hotel. The agent reads the request and shortlists options. Then control comes back to Dani to confirm, change, or cancel, with Face ID over 100 dollars. Finally, the agent follows through, steps in if something looks unsafe, and hands control back.

PROTOTYPE

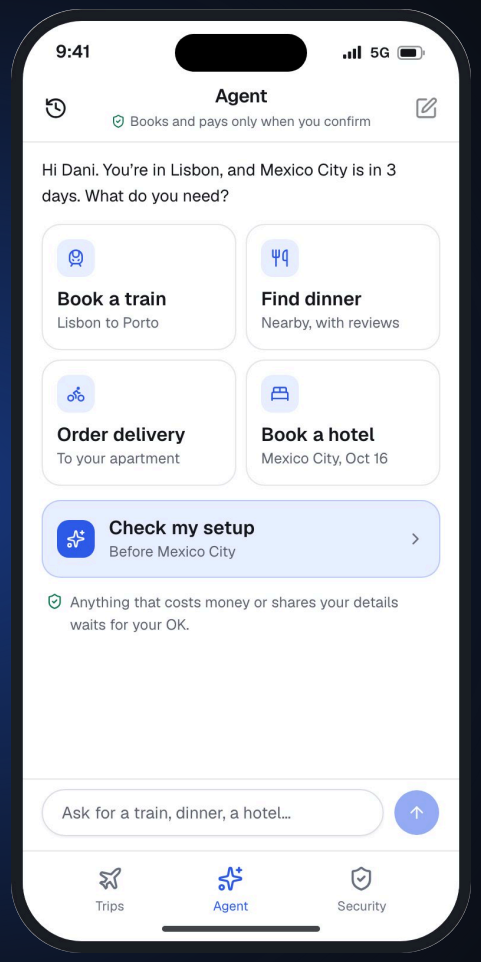
Live demo.
Four moments to watch.

Book a hotel
Confirm card, then Face ID

Find dinner
A transfer deposit, refused

Cancel a booking
The refund, spelled out

Check my setup
Explain, approve, undo



SPEAKER NOTES · 08 LIVE DEMO

Now let's see it live. First, I'll book a hotel. Here's the confirm card, and because it's over 100 dollars, it asks for Face ID. Next, dinner. This restaurant wants a deposit by bank transfer, so the agent refuses and says why. Then I'll cancel a booking, and it shows exactly what comes back. Finally, Check my setup explains each step, and nothing changes until I approve it.

HUMAN FLOURISHING

Agency and learning.
Help that doesn't take over.

	Opportunity How it helps		Risk How it could backfire		Safeguard What we built
Agency Does Dani stay in control?	Dani makes every choice that matters.	but →	Dani may tap Confirm without reading.	so →	Confirm cards show only cost, payment, and what's shared. Face ID only over $100.
Learning Does Dani learn to do it alone?	The agent explains every warning, so Dani learns to spot scams.	but →	Dani may lean on the agent and stop learning.	so →	Every setup step shows how to do it yourself next time.

THE TENSION

Hiding scam listings protects Dani but removes choices. So the agent always says what it left out.

NEXT TEST

Five nomads try it while a teammate plays the agent. Do they read the confirm card? Can they repeat a setup step alone?

SPEAKER NOTES · 09 HUMAN FLOURISHING

For agency, Dani makes every choice that matters. But Dani might tap Confirm without reading, so confirm cards show only cost, payment, and what's shared, and Face ID only appears over 100 dollars. For learning, the agent explains every warning, so Dani learns to spot scams. But Dani might lean on it, so every setup step shows how to do it yourself. And when it hides a scam listing, it always says so. Thank you.

APPENDIX A

Where the AI enters, and where it stops.

What activates it?
Dani opens the Agent tab, or taps Check my setup after a profile downloads. Nothing runs in the background.

Does it ask, or suggest?
Dani asks. Its one suggestion: after a hotel booking, it points to setup fixes, and Dani decides whether to open them.

What information does it use?
The request, trip city and dates, the card on file, and network status. The setup check sees only what Dani shares.

Does Dani know what it uses?
Yes. The share screen lists each source, and every confirm card says what gets shared, and with whom.

What does it produce?
Shortlists, risk flags with reasons, confirm cards, bookings in Trips, and a setup checklist.

What needs approval?
Anything that costs money or shares personal details. Over $100 also needs Face ID.

Can Dani edit, reject, or reverse it?
Change or Cancel on every card. Bookings cancel from Trips. Setup approvals can be undone, except a sent Keyholder card.

How does it show uncertainty?
Prices are about-amounts in dollars, Wi-Fi speeds come from other nomads' tests, and ratings show review counts.

What if it can't finish?
It says so, explains why, and offers another way: a different option, or the steps to do it yourself.

When does it stop?
After the receipt and one next step. It never chains bookings on its own.

How does control return?
The booking sits in Trips, the chat waits, and every thread is saved in history.

SPEAKER NOTES · 10 APPENDIX A: PLACING THE AI

If you're wondering exactly where the AI enters: it only starts when Dani opens the Agent tab or taps Check my setup. It uses only what Dani shares, anything that costs money or shares details needs approval, and every booking can be changed or cancelled.

Opportunity statements and value proposition.

AREA	OPPORTUNITY STATEMENT	HOW MIGHT WE
Network judgment	Members value **competence** and participate through **VPN and travel-router routines**, but experience tension between **fear-driven advice** and **unclear real risk**.	How might we help nomads judge network risk with confidence without reinforcing fear-driven tool buying?
Security that moves with you Selected	Members value **self-reliance and freedom of movement** and participate through **second phones, two numbers, and pre-departure prep**, but experience tension between **security systems that assume a home** and **lives spent crossing borders**.	How might we help nomads arrive in any country already set up, and recover from anywhere, without assuming a home or hiding where they are?
Money that keeps moving	Members value **freedom of movement** and participate through **multi-bank, multi-app money stacks**, but experience tension between **automated compliance** and **travel patterns that look suspicious**.	How might we help nomads keep access to money during a freeze without needing a home bank to fall back on?

VALUE PROPOSITION

For **digital nomads and remote workers** who value **self-reliance and freedom of movement** but experience **security systems that assume a home**, NOMAD.OS helps them **arrive in any country already set up, and recover from anywhere** by **downloading one profile per country that sets four layers: network, money, numbers, and a safety net**.

SPEAKER NOTES · 11 APPENDIX B: OPPORTUNITY STATEMENTS

These are the three opportunity areas from our Assignment 4 research. We chose security that moves with you: helping nomads arrive set up and recover from anywhere. This agent grows out of that choice.

Evidence table, 1 of 3 E1 to E9

ID	SOURCE	OBSERVATION	INTERPRETATION	VALUE	TENSION
E1	Creator setup guides, Jul and Oct 2026	Gear-by-gear stacks: a named travel router running WireGuard, multiple eSIMs, priced and weighed per layer.	The setup itself is a display of competence.	Preparedness, control	Preparedness vs. burden and cost
E2	Nomad tools blog, Sep 2026	Argues VPN value depends on the work; light, non-sensitive work on a personal hotspot may not need one.	Nuance entering a VPN-maximal conversation.	Practicality	Fear vs. actual need
E3	VPN vendor blog, Aug 2026	Opens by saying most public Wi-Fi guides sell fear or sell a VPN; calls a VPN often overkill for low-risk use.	Even sellers distance themselves from fear marketing.	Calm competence	Marketing vs. trust
E4	VPN vendor blog, two posts, Jul and Aug 2026	One post says HTTPS protects you from hackers on public Wi-Fi; the other says HTTPS alone won't keep you safe.	Guidance is unstable even within one source.	Clarity	Conflicting advice
E5	VPN blog, Jul 31, 2026; travel blog comments, 2026	Readers describe buying travel routers so an employer sees a home IP; the vendor warns detection is likely.	Security gear doubles as concealment.	Freedom	Protection vs. concealment
E6	Travel blog post and reader comment, Jul 2026	A perpetual traveler's phone is stolen; a reader notes phones are grabbed while unlocked in use.	Risk peaks at the moment of use, not at rest.	Vigilance	Convenience vs. safety
E7	Phone accessory brand blog, Aug 2026	Recommends carrying a secondary "burner" phone for risky situations abroad.	Redundancy extends to whole devices.	Preparedness	More gear vs. simplicity
E8	Hacker News, Ask HN, Sep 2026 (not travel-specific)	After a phone theft, the poster can't reach banking, email, or cloud files. Replies: keep two signed-in devices; keep codes in a password manager.	One device is a single point of failure; peers prescribe redundancy.	Self-reliance, peer help	Security vs. access
E9	Q&A help site, Oct 2026 (traveler, not confirmed remote worker)	Traveler in New Zealand is locked out of Gmail; her recovery number is an Australian line she can't use.	Recovery paths assume you're near home.	Access	Mobility vs. verification

SPEAKER NOTES · 12 APPENDIX C: EVIDENCE TABLE, 1 OF 3

This is the evidence behind the codes on slide 3: 27 public posts from our Assignment 4 research. For each one, we noted the source, what we saw, and what we think it means.

Evidence table, 2 of 3 E10 to E18

ID	SOURCE	OBSERVATION	INTERPRETATION	VALUE	TENSION
E10	Google Account Help, current page (context)	Recovery tips: use a familiar device and a location where you usually sign in, like home or work.	Platform policy encodes a home.	n/a	Home assumption
E11	Apple Support, current page (context)	Stolen Device Protection adds biometric checks and a delay when the phone is away from familiar locations.	The same assumption, in theft protection.	n/a	Home assumption
E12	r/digitalnomad thread, Apr 22, 2026 (also summarized in May 2026)	About $68k held by a fintech app for nearly 6 months; 230+ comments. Replies: it isn't a bank, keep a real one; regulator complaints released funds. Update: money returned late May.	Failure stories become community rules.	Collective knowledge	Convenience vs. control of money
E13	Expat VPN site, Oct 2026; nomad tax guide, Jun 2026	One says a dedicated-IP VPN ends repeated selfie checks; the other says VPNs can trigger fraud flags and freezes.	The same tool is sold as cure and blamed as cause.	Access	Conflicting advice
E14	Expat phone setup guide, May 2026	Banks and government portals send codes only to real US numbers, mostly not VoIP; recommends a cheap carrier line kept active from abroad.	Nomads rebuild a "home" number.	Continuity	Freedom vs. home dependence
E15	Nomad blog, Apr 2026 (8 years nomadic)	Calls the password manager the keystone; keeps a printed emergency kit sealed with a trusted person at home; warns against carrying the only copy.	Recovery depends on people and place.	Care, preparedness	Freedom vs. home dependence
E16	r/digitalnomad, ~Jun 2026, 70+ comments	PSA that the phone is the key to everything; replies advise storing IMEI and carrier details in a password manager; one reply also locked out of Google.	The phone is a single point of failure, and the community knows it.	Preparedness	Security vs. access
E17	r/digitalnomad, ~Aug 2026, 40+ comments	Thread on carrying a secondary phone; the top answer adopted it after a phone snatch in Barcelona.	Redundancy is learned through failure.	Preparedness	More gear vs. simplicity
E18	r/digitalnomad, May to Oct 2026, three threads	Full-time travel "ruined" a poster's number; replies split life across two numbers; the keep-your-US-number question recurs and draws weary replies.	The home number is infrastructure nomads must maintain from afar.	Continuity	Freedom vs. home dependence

SPEAKER NOTES · 13 APPENDIX C: EVIDENCE TABLE, 2 OF 3

These items include the platform policies that assume you have a home, and the threads about keeping a home number alive for bank codes.

APPENDIX C · FROM ASSIGNMENT 4

Evidence table, 3 of 3 E19 to E27

ID	SOURCE	OBSERVATION	INTERPRETATION	VALUE	TENSION
E19	r/digitalnomad, ~May 2026, 160+ comments	A newcomer about to quit and travel is told to carry two phones and absolutely keep a home number for bank and government codes.	Status is passed down as rules for newcomers.	Belonging, care	Freedom vs. home dependence
E20	r/digitalnomad, Apr to Aug 2026, 120 to 390+ comments	Stealth-nomading threads reference the subreddit wiki setup (travel router, tunnel home, kill switch); a reply says IT reads a foreign login as stolen credentials.	Concealment is organized and documented, not fringe; corporate security also assumes a home.	Freedom, control	Protection vs. concealment
E21	r/digitalnomad, ~May 2026, 10 to 20+ comments	Newcomers ask whether a $10k transfer will trigger a freeze, and report transfers held for extra security checks.	Failure stories shape behavior before failure happens.	Control of money	Convenience vs. control
E22	X, Sep 21, 2026, creator (~23.5k followers), 367 likes	Phone stolen by motorbike thieves in Asunción; cards added to Apple Pay and thousands taken; 2FA tied to the stolen phone and a Spanish SIM, so he flew to Spain to recover the number.	Recovery can require physically returning to a “home” country.	Self-reliance	Freedom vs. home dependence
E23	X, Oct 5 to 8, 2026, three small accounts	One says get your cell number out of 2FA and use Google Voice; one says banks may refuse Voice numbers, so test each bank; one says keep the home SIM on just for codes.	Advice conflicts in the same week, with no shared answer.	Clarity	Conflicting advice
E24	X, Oct 6, 2026, developer creator (~400k)	Praises a GL.iNet travel router for making travel easier; a VPN is optional.	The router is also a mainstream convenience gadget, not only a security or concealment tool.	Convenience	(Complicates the burden tension)
E25	X, Oct 1, 2026, small nomad account	Describes briefing an eSIM provider on 2FA, eSIM, and the financial risks of a stolen phone.	The market is starting to notice the same gap.	n/a	Competitive signal
E26	r/digitalnomad, ~May 27, 2026, 38 upvotes, 39 comments	A work account was frozen after using a consumer VPN while traveling; replies say commercial VPNs are easy to detect and suggest self-hosting at a parent’s house.	The security tool triggers the lockout, and the fix routes back through home.	Access	Security vs. access
E27	r/GlInet, ~Apr 8, 2026, 13 comments (verified via a listing)	A newcomer offers to pay for help setting up a travel router to present a home IP; replies cover home-side servers, 2FA, and disabling Wi-Fi on work laptops.	Concealment setups are hard enough that newcomers pay for help.	Freedom	Protection vs. concealment

SPEAKER NOTES · 14 APPENDIX C: EVIDENCE TABLE, 3 OF 3

And these include the stolen phone in Asunción, and three conflicting answers about bank codes in a single week.

APPENDIX D · FROM ASSIGNMENT 4

Research-to-design traceability.

EVIDENCE	WHAT WE OBSERVED OR CONCLUDED	DESIGN RESPONSE
E10 E11 E22	Recovery keys off home; one nomad flew home to get a number back.	Safety net: Break Glass from any browser; a Keyholder replaces the familiar-place check. (S6, S7)
E13 E23 E26	The right answer depends on the country and the bank.	Country profiles, with dated reports for each country. (S1, S2)
E2 E3 E4	VPNs: needed versus overkill, and fear-driven advice.	Network layer: VPN on only for untrusted Wi-Fi. (S3)
E13 E26	VPNs trigger bank and work-account freezes.	Network layer: apps that flag VPNs go direct. (S3)
E6 E22	Thieves add cards to wallets and move money fast.	Money layer: cards locked to trip dates and places. (S4)
E12 E21	Fintech freezes strand money for months.	Money layer: a backup way to pay and a freeze plan. (S4)
E14 E18 E19	Nomads keep a home number alive for codes.	Numbers layer: home line for codes, local eSIM for data. (S5)
E8 E23	Advice splits on SMS versus other code methods.	Numbers layer: codes that can leave SMS move to an authenticator. (S5)
E15 E17	Nomads leave kits with family and add backups after a theft.	Safety net: a Keyholder with a split key. (S6)
E5 E20	Security gear is used to hide location from employers.	Principle: encrypt, don’t disguise. The VPN exits where you are. (Principles)

SPEAKER NOTES · 15 APPENDIX D: TRACEABILITY

This table traces each finding to a design response, and to the screen in the prototype that carries it.

APPENDIX E · FROM ASSIGNMENT 4

Sources for every evidence item.

E1 dev.to/notifyshow/the-ultimate-digital-nomad-setup-minimalist-tech-and-working-from-anywhere-179k; nomadstoolslab.com/digital-nomad-setup-guide-2026

E2 ontheroadtools.com/vpn-for-digital-nomads-do-you-actually-need-one

E3 windscribe.com/blog/how-to-use-public-wi-fi-safely

E4 protonvpn.com/blog/vpn-myths; protonvpn.com/blog/public-wifi-and-https

E5 vpn.pro/blog/can-your-employer-detect-that-youre-using-a-vpn-while-working-remotely; indietraveller.co/best-travel-vpn (comments)

E6 johnnyjet.com/dont-fall-for-this-new-phone-theft-scam-targeting-tourists

E7 phoneloops.com/blogs/posts/phone-security-while-traveling-europe-prevent-theft-stay-safe-in-2026

E8 news.ycombinator.com/item?id=49742976

E9 justanswer.com/email/jeery-locked-gmail-account-currently-overseas-none.html

E10 support.google.com/accounts/answer/7299973

E11 support.apple.com/120340

E12 reddit.com/r/digitalnomad/comments/1ssjy5m; summary at moveandmoney.com/blog-fintech-account-frozen-2026

E13 safeexpatvpn.com/revolut-account-restricted-abroad; nomadtaxguide.com/blog/digital-nomad-banking-guide

E14 expatretireguide.com/money/phone-setup

E15 tonydehnke.com/blog/digital-nomad-backup-tools

E16 reddit.com/r/digitalnomad/comments/1uh3o4f

E17 reddit.com/r/digitalnomad/comments/1vwj8pg

E18 reddit.com/r/digitalnomad/comments/1w75mkf; 1wyr9jc; 1t8q1ig

E19 reddit.com/r/digitalnomad/comments/1ti45hs

E20 reddit.com/r/digitalnomad/comments/1vzmfww; 1tr5hou; 1slwlzs

E21 reddit.com/r/digitalnomad/comments/1tqfsq4; 1tlf6pc

E22 x.com/hispanicnomad/status/2102094776187134436

E23 x.com/GabeLenz/status/2108298536093896991; x.com/StephenZhou1004/status/2108115835294216253; x.com/nexkit2026/status/2107078468936908918

E24 x.com/theo/status/2107590392128704776

E25 x.com/baddigitalnomad/status/2105673102768419026

E26 reddit.com/r/digitalnomad/comments/1tp3a7b

E27 reddit.com/r/GlInet/comments/1sfksl9

Reddit items come from Google listings filtered to the study window, dates estimated from post IDs; X posts were opened and checked by hand. Dates are publication or comment dates where shown; otherwise estimated from the page's indexed age and given to the month. Policy pages were read in October 2026. Usernames are omitted throughout.

SPEAKER NOTES · 16 APPENDIX E: EVIDENCE SOURCES

Here are the sources for every evidence item, and how we collected and dated them.

APPENDIX F

References.

Airalo. (2026, June 2). *Airalo surpasses 30 million users* [Press release]. https://www.airalo.com/about-us/press-releases/airalo-surpasses-30-million-users	Existing product analysis
Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. *Proceedings of CHI 2019*. https://doi.org/10.1145/3290605.3300233	Explain, confirm, and support correction
Google PAIR. (2021). *People + AI guidebook*. https://pair.withgoogle.com/guidebook	Trust, feedback, and errors
OpenAI. (2025, January 23). *Introducing Operator*. https://openai.com/blog/introducing-operator	Precedent: an agent that asks before it buys
Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics, Part A, 30*(3), 286-297. https://doi.org/10.1109/3468.844354	How much the agent does on its own
Shneiderman, B. (2022). *Human-centered AI*. Oxford University Press.	High automation with high human control
Federal Trade Commission. (n.d.). *Avoid scams when you travel*. Consumer Advice. https://consumer.ftc.gov/articles/avoid-scams-when-you-travel	Wire transfers as a scam signal
Airalo. (2026). *airalo.com, mobile home page* [Screenshot]. Captured October 2026.	Phone on slide 2
Unsplash. (n.d.). *Traveler with a backpack* [Photograph]. Unsplash License. https://images.unsplash.com/photo-1606525380696-c2bfe72f7b8d	Photo of Dani on slide 3
NOMAD.OS team. (2026). *Assignment 4 research: 27 public posts on how nomads stay secure, E1 to E27*. IDSN 523.	Evidence codes on slide 3, in full in Appendix C

SPEAKER NOTES · 17 APPENDIX F: REFERENCES

And these are our references: the products, precedents, and research we drew on.